

Human Fall Detection in Surveillance Videos

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Abstract—The development of healthy aging and the assistance of older people in their everyday activities is one of the most pressing concerns in modern cultures. Human fall is such an irregular action that occurs when a person gets startled and loses their equilibrium, resulting in sliding. These accidents among old people can result in long-term impairments and even death. Automated detection of falls has gotten a lot of interest in computer vision and pattern recognition areas because of its social and economic implications. Although wearable sensor-based techniques have achieved excellent detection rates, some potential users are hesitant to wear them, and so their usage has not yet become standard. As a result, new ways have arisen, such as vision-based solutions. Fall detection that is accurate for the help of older persons is critical in reducing the number of fatalities and injuries caused by falls. The influence of deep learning on vision-based systems, such as action recognition, has transformed the landscape. The emergence of Smart Environments and Internet of Things concepts, together with the growing number of cameras in our everyday lives, creates an ideal setting for vision-based solutions. As a result, this paper aims to provide a vision-based method that uses Convolutional Neural Networks to determine if a series of frames involves a fall event or not. The proposed approach is using Histogram of Oriented Gradient (HOG) features extracted from series of frame to determine the fall event.

Index Terms—Deep Learning, Computer Vision, Histogram of Oriented Gradient, Pattern Recognition.

I. INTRODUCTION

The population is aging at a quicker rate than ever before. According to the United Nations Population Fund (UNFPA) (UNFPA, 2012), by 2050, the worldwide population of adults aged 60 and up would reach 2 billion. Currently, health related problems have targeted a large percentage of older population. Many people are facing problems in living because of these challenges, which are compounded by the normal decrease in physical and cognitive abilities of the aged. Due to the physical weakness associated with aging, fall occurs frequently among old people, and it has bad impact on their health by making them prone to injury resulting in health issues. In HMS (Health Monitoring Systems), rapid and accurate fall detection is critical for providing appropriate services and medical assistance [1]. Thus, human fall is an alarming event which must be monitored. Surveillance video cameras give crucial information for watching people's behaviour throughout the fall event. Various approaches employing modern technologies such as wearable sensors, accelerometers, gyroscopes, magnetometers, and so on have been presented in recent years

to determine the correct fall. However, wearing a device for an extended period of time is impractical for the elderly, this is not an effective option. As a result, it is critical to implant a penetrating monitoring system for elderly citizens that can identify fall behaviours within the room promptly and automatically which alerts caregivers to the situation. For this purpose, a sensor-based alarm production system is needed to be installed in the living room. In such scenario, as vision based system can be used for health mentoring setup.

The core objective of this paper is to devise a vision-based solution for the detection of human falls by making use of a sequence of image frames in the streaming surveillance videos. It aims to develop a comparatively faster and more accurate solution whose usage can be encouraged in real-life scenarios for the betterment of elderly people. The rest of this paper consists of following sections, Section II discusses the literature review for fall detection problem. Section III puts forward the proposed Histogram of Oriented Gradients (HOG) [13][17] based solution. Section IV discusses and compares the results produced by proposed model and optical flow [16] based model presented in base paper [2]. Section V consists of conclusion and future work.

II. LITERATURE REVIEW

Typical fall detection techniques which have been used in this field are sensor based and vision based. Sensor-based detection has traditionally relied on accelerometers, which offer accurate acceleration measurements such as vertical acceleration. These measurements are extremely distinct from daily activities in the case of falls, allowing us to distinguish between them. Vision based techniques rely on human motion captured in video sequences and distinguish activities using feature based classification.

A. Sensor Based

Rungnapakan and Chintakovid [9] proposed a fall detection technique making use of accelerometer and gyroscope sensors installed in user smartphone. This approach is based on the change in acceleration and orientation to detect fall events. Whenever there will be abrupt change and threshold breach in the sensor values fall event can be detected. The disadvantage of sensor-based approach is that there may be false alarms caused due to non-fall events such as accidental dropping of

mobile phone. User may not have mobile phone with him/her all the time, so some injurious fall may not be detected.

Yuxi Wang, Kaishun Wu and Zhong Zhang, Christopher [10] [11] devised fall detection techniques without using any handheld or wearable sensors by installing the tracking system in the surrounding of people. Such techniques have eliminated the need for wearable or handheld sensors for which people may be reluctant to wear or carry but such technology cannot be used on the go and is limited to some specific arena.

B. Vision Based

Sagar Chhetri, Abeer Alsadoon and Núñez-Marcos, Azkune [2] [3] had developed a technique for fall detection using the optical flow in image frames of video sequences. The optical flow features were extracted and were refined using a deep convolutional neural network. The finally extracted features were passed to classifier layers composed of a fully connected followed by a softmax layer. This approach concluded the fall detection with accuracy of 93%. The main challenge posed by this approach is very high processing power and time required for generating optical flow features on images.

Kun Wang, Guitao Cao [4] had devised a vision-based fall detection technique by combining computer vision-based features, namely, Histograms of Oriented Gradients (HOG), Local Binary Pattern (LBP) and a feature extracted by the Deep Learning Framework Caffe to form a combination a feature set. Each image frame was pre-processed to extract these features and combination feature vector is generated on each frame. Classification is done on each frame using SVM classifier, thus it worked as a single frame classification model. This work has reported the sensitivity of 92% and specificity of 89%. This approach is facing challenges because of its single frame classification nature instead of working on a sequence of frames, which led to compromised accuracy of the model.

F. Harrou, N. Zerrouki [5] used tracking techniques in a near area to detect falls. Authors proposed a methodology for computing a person's silhouette and extracting information such as the spatial orientation of the silhouette's centre orbits a geometric orientation using labelling of connected components.

The use of 3D vision to take advantage of 3D structures is another area of visionbased fall detection systems. This technique necessitates the use of many cameras (passive systems) or active depth cameras, such as the Microsoft Kinect or time-of-flight cameras, to derive depth maps. Because of its inexpensive price and outstanding performance, the Kinect camera is extremely popular. Auvinet et al. [6] employed a Kinect camera to create a 3D silhouette, which was then used to examine the volume distribution along the vertical axis. Gasparrini et al. [7] utilised a similar camera to extract 3D data before using a tracking system to identify falls. Kinect software offers body joints, which Planinc and Kampel [8] utilised to determine the main axes' orientation based on their location. The disadvantages of such methods are connected to system deployment: they require either several synchronized

cameras focused on the same region or active depth cameras with small fields of vision and limited depth.

Vision-based solutions are preferred because surveillance video cameras are prominent these days but they require high processing power. The sensor-based solutions have the disadvantage of giving confounding results due to the changed orientation of sensors while activity detection, therefore a fixed reference of orientation has to be maintained while activity detection. Also, users may be reluctant to wear or hold sensors, which puts a limit on sensor-based solutions. In the purposed work, a vision-based solution will be developed.

III. THE PROPOSED APPROACH

A. Overview

In the proposed approach, there are mainly two components which work on a sequence of frames extracted from videos: feature extractor and classifier. The output of the classifier will result in probability of fall/non-fall class. All the steps are briefly described in following text.

a) *Reading Input Video*: The sample video is read frame by frame, videos can be either MP4 or AVI. The RGB frames extracted from the videos will have 3 channels

b) *Frame Extraction and Pre-processing*: Each RGB frame will be read in sequence and pre-processing of each frame will be done. The frames will be resized to a uniform shape of 224x224, because it is the minimum possible size up to which image dimensions can be reduced without losing image statistics [12]. Pre-feature extraction is performed using histogram of oriented gradients (HOG) feature descriptor used in computer vision and image processing. Finally, the image sequence will be divided into image stacks.

c) *Feature Extractor*: Each image stack of size 20 will be fed to feature extractor network for refining frame features and extract final features based on the change/motion in the image, such as required action can be recognized. Feature extractor will be a deep convolutional neural network architecture, so that it can extract a good amount of image features.

d) *Feature Classifier*: Feature classifier will receive the feature vector obtained as the output of previous step. The classification network will detect if the sequence of frames in a stack involved a fall event or not by delineating the probability of each class, namely, a fall and non-fall. The class that has higher probability will be taken final output of the model.

B. Methodology

The methodology used in the proposed solution is mainly based on Histogram of Oriented Gradients (HOG) features of frame sequences. This method is also compared with existing Optical Flow based features[2][3]. HOG features are popularly used in computer vision based techniques which are widely used in the field of object detection. In this work, this technique is first time applied to the series of frame to detect the motion and consequently extract features which are useful for Human activity recognition in contrast to existing work where Histogram of Oriented Gradient (HOG) is used in single frame-based classification model along with Local Binary

Pattern (LBP) [4]. Histogram of Oriented Gradient (HOG) features have been chosen because of their ability to detect change in orientation and direction in the consecutive image frames. Also, HOG feature extraction is simpler as compared to optical flow feature extraction [2][3] while ensuring good accuracy. HOG features focus on change in orientation and gradient in the series of frames to detect motion. The change in the consecutive RGB frames during a sample fall sequence and their corresponding HOG frames are shown in “Fig. 1”. In the proposed method firstly computer vision based algorithms are applied on the sequence of frames. These algorithms consist of resizing the image frames to uniform size, RGB to grayscale conversion of image and histogram of oriented gradients feature extraction. After performing these pre-processing steps on the series of frames deep learning based algorithms are used for feature and extraction and final classification of image sequences into fall/non-fall categories. This paper purposes a computer vision based solution for fall detection. The model design represented in “Fig. 2” is described in detail in the following text.

a) *Reading Model Input:* All the videos in datasets were downloaded in a repository. Each video is converted into frames using functions provided in OpenCV. Each frame is then resized to 224x224 and then gray-scaled. The series of frames in each video is then read stack by stack with the stack size of 20 and a sliding window of size 1. Each stack will act as a single instance of model input, that is, input of size 224x224x20.

b) *Pre-processing of Input Frame stack:* Before passing the input to the neural network model, the histogram of oriented gradient is generated on each frame to perform another level of pre-processing. It will highlight the shape of the object in the image, remove unwanted information from the frame. In the base paper model [2] the optical flow features are extracted at this stage instead of the histogram of oriented gradients. Optical Flow requires high processing as compared to the histogram of oriented gradients since optical flow calculation involves all the 3 channels of the frame, RGB(Red, Green and Blue) for the computation whereas histogram of oriented gradients can be calculated using a single grayscale channel only. Now the input is ready to be passed through the neural network layers for feature extraction and classification.

c) *Feature Extractor:* Now the video sequences will be passed through the feature extractor to refine the features. For the purpose of feature extractor, generic VGG-16 [12] deep convolutional neural network architecture is used because has shown promising results with other human activity recognition datasets. Multiple convolution layers with different number of filters (64, 128 and 512) have been used. Each filter has a size of 3x3. The layers are arranged in the increasing order of filter size and max pool layers in between are helping in reducing dimensions of input when the input image is going through a set of convolutional layers. Detailed feature extractor is shown in “Table I” and is summarized in the following text.

- Input: It is a stack of HOG images of size 224x224.
- Output: It is a feature vector of size 4096.

TABLE I: Feature Extractor

Neural Network Layer	Output Size
Convolution(filters=64,kernel_size=3x3)	224x224x64
Convolution(filters=64,kernel_size=3x3)	224x224x64
Maxpool(pooling_size=2x2)	112x112x64
Convolution(filters=128,kernel_size=3x3)	112x112x128
Convolution(filters=128,kernel_size=3x3)	112x112x128
Maxpool(pooling_size=2x2)	56x56x64
Convolution(filters=256,kernel_size=3x3)	56x56x256
Convolution(filters=256,kernel_size=3x3)	56x56x256
Convolution(filters=256,kernel_size=3x3)	56x56x256
Maxpool(pooling_size=2x2)	28x28x256
Convolution(filters=512,kernel_size=3x3)	28x28x512
Convolution(filters=512,kernel_size=3x3)	28x28x512
Convolution(filters=512,kernel_size=3x3)	28x28x512
Maxpool(pooling_size=2x2)	14x14x512
Convolution(filters=512,kernel_size=3x3)	14x14x512
Convolution(filters=512,kernel_size=3x3)	14x14x512
Convolution(filters=512,kernel_size=3x3)	14x14x512
Maxpool(pooling_size=2x2)	7x7x512
Flatten	25088
Dense(to=4086)	4086

- Choice of the number of filters: The number of filters in each layer will be the same as the VGG-16 network because of its accuracy with other activity recognition datasets.
- Choice of filter size: It will be 3x3, which is the minimum possible filter size to recognize motion in all 4 directions.
- Choice of max-pool size: It will be 2x2, which is the minimum possible size to reduce dimensions with minimum loss.

d) *Feature Classification:* The feature vector of size 4096 is obtained for each frame. Each instance will contain stack of 20 feature vectors of consecutive frames, then passed through fully-connected and softmax layers to get the classification probability for each class.

- First Phase (Model Training): In the first phase, all the feature vectors corresponding to each training video will be used to fit the classifier model along with the ground truth, fall or non-fall video. In this way, the model will learn the features corresponding to both the events. Since ground truth is provided to the model during training, it is a supervised learning algorithm.
- Second Phase (Model testing): Once the model is trained, test samples can be passed to the classifier. The sample will be classified by the model into fall and non-fall categories. The output layer in the model will be softmax activation which outputs the probability with which sample belongs to the non-fall category. Say the output of the model is P, then
 - 1) Probability of non-fall event (P_1) = P
 - 2) Probability of fall event (P_2) = 1-P
 If $P_2 > P_1$ or $P < 0.5$, the sample will be classified as fall else the sample will be classified as a non-fall event.

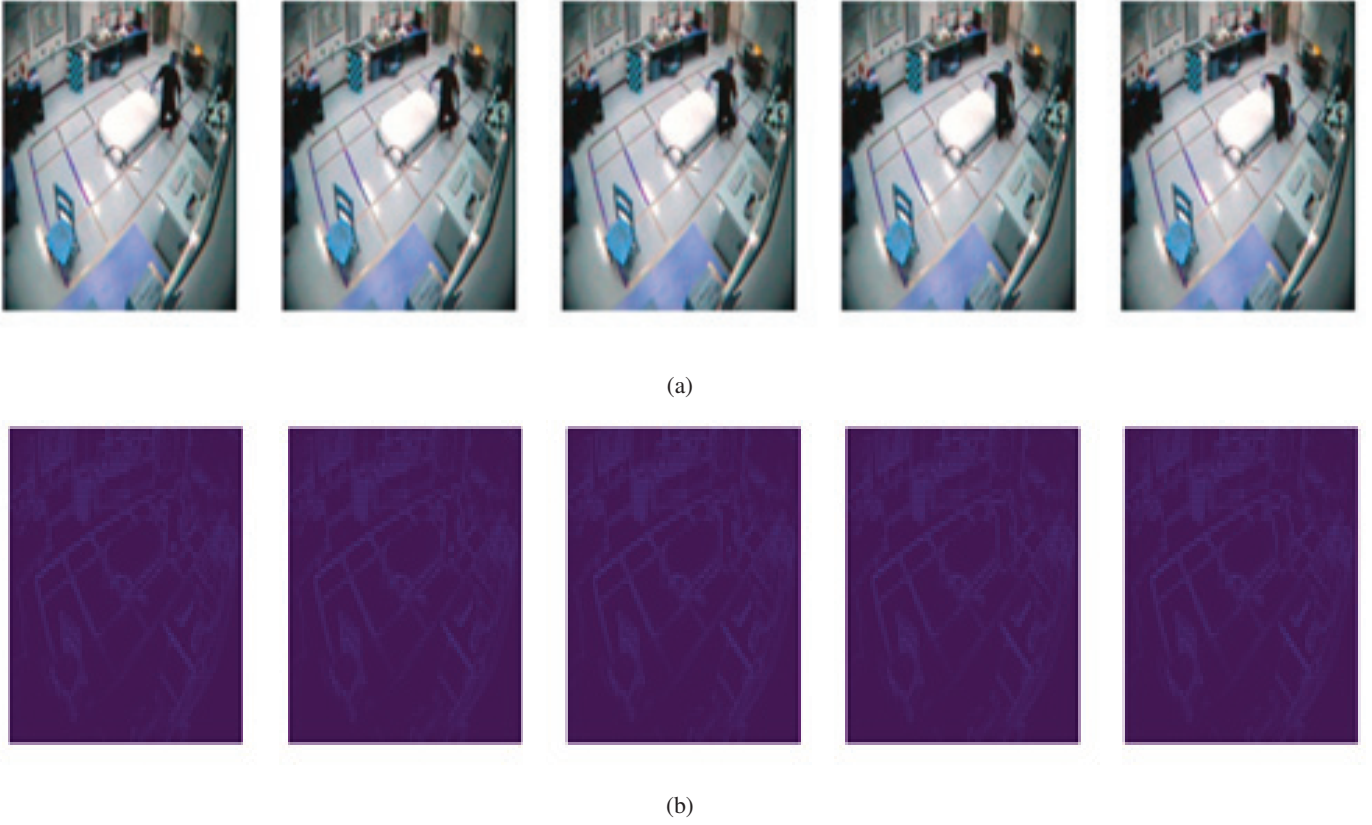


Fig. 1: Consecutive (a) RGB and (b) HOG frames (left to right) displaying a person tending to fall

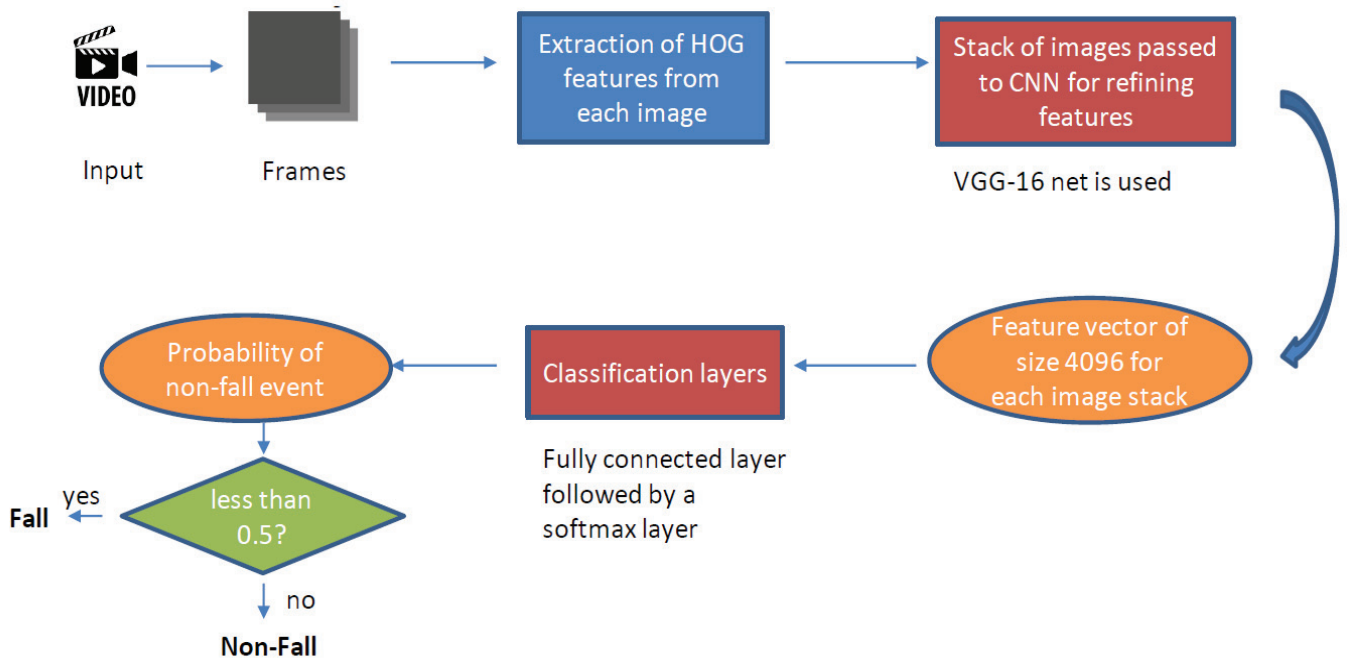


Fig. 2: The model takes video as input, extract raw RGB frames from video and then extract HOG features for each frame in sequence. These pre-processed frames are passed to deep neural network feature extractor and stacked in the size of 20 for performing final classification. Each stack of 20 frames along with their extracted features will result in a fall or non-fall class.

C. Dataset

In this work two publicly used datasets are used, namely, UR Fall Detection Dataset and Multiple Cameras Fall Dataset. The description of these two datasets is as follows:

- UR Fall Detection Dataset [14]: This dataset has total 70 videos comprising of 30 video sequences of fall and 40 video sequences of activities of daily life (non-fall). The frame size of each frame in video is 224x224. All the fall videos are captured in the same room but changing background in terms of placement of objects in video frames. All the non-fall videos are captured in 5 different rooms thus different backgrounds. Different persons are involved in capturing fall and non-fall samples. Along with RGB video sequences, depth information of the video sequences is also available. All the videos are available in MP4 downloadable format.
- Multiple Cameras Dataset[15]: This dataset contains total 22 fall videos having 22 fall scenarios and 2 non-fall videos having 20 scenarios captured using 8 IP cameras. The frame size of each frame in video is 480x480. The falls are captured from different angles in the room. All the falls in this dataset are captured in same but recorded from different angles. Various scenarios are captured such as falling from front, back and sideways. Falls in this dataset are useful in training the model for different fall scenarios in terms of fall postures and orientation. All the videos are available in AVI format and zipped in a single downloadable zip file.

Total fall scenarios: $22 \times 8 = 178$

Total non-fall scenarios: $20 \times 8 = 160$

a) *Experimental Setup*: The project work was simulated using google colab python 3.7 notebook along with GPU. GPU is used for training and testing of proposed models because of high processing power required for processing images and feature extraction, classification using deep neural network. The GPU has accelerated the processing thus helped in generating the results faster. The system configuration of an online google colab platform on which experiments are done has been listed in “Table II”.

TABLE II: System Configuration

Type	Configuration
Processor	Intel Xenon
Speed	2.3 GHz
RAM Size	13 GB
Number of Processors	2
GPU size	K80
GPU Memory	12 GB
Configuration	64 bit

IV. RESULTS

a) *Input Pre-processing Results*: In the proposed model after frame extraction, the pre-processing involves resizing of the frame to uniform size of 224x224. The image is then converted to grayscale since HOG features can be extracted

from just one gray-scaled channel. HOG extraction algorithm is applied to the gray-scaled frame. The various stages of frame are shown in “Fig. 3”.

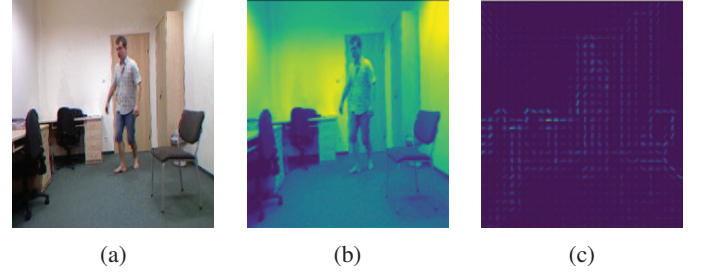


Fig. 3: Raw RGB frame extracted from video is shown in (a). It has 3 channels but for simplicity these 3 channels can be merged to 1 channel before processing the frame further. Therefore, RGB frame is converted to Grayscale frame which is shown in (b). The Grayscale frame has to be processed to obtain HOG frame. The HOG frame obtained from Grayscale frame is shown in (c). HOG frame will be the final result of pre-processing which will be passed to next stages of the model.

b) *Model Testing Results*: The proposed model and model presented in the base paper[2] are implemented and compared based on sensitivity, specificity, and accuracy metrics for UR Fall dataset [14] and Multiple cameras dataset [15]. The results obtained for these two datasets are shown in “Table III”. Definition of each of the metric used for evaluating the model is given below:

- Sensitivity: (Number of true positive assessment)/(Number of all positive assessment)
- Specificity: (Number of true negative assessment)/(Number of all negative assessment)
- Accuracy: (Number of correct assessments)/(Number of all assessments)

TABLE III: Model Performance

Metric	UR Fall Dataset		Multiple Cameras Dataset	
	HOG model	Optical Flow Model [2]	HOG model	Optical Flow Model [2]
Sensitivity	100%	100%	94.5%	94.5%
Specificity	87.5%	87.5%	90.9%	90.6%
Accuracy	92.8%	92.8%	97.4%	96.1%

Classification details on test datasets using random split of 80:20 is presented in “Table IV”. From the results, it can be

TABLE IV: Classification Results

	UR Fall Dataset		Multiple Cameras Dataset	
	HOG model	Optical Flow model [2]	HOG model	Optical Flow model [2]
Total scenarios	14	14	77	77
Fall scenarios	6	6	36	36
Non-fall scenarios	8	8	31	31
True positives	6	6	35	35
True negatives	7	7	30	29
False positives	1	1	1	2
False negatives	0	0	1	1

seen that both HOG model and Optical flow based model have shown similar results for UR Fall dataset, both have shown

accuracy of 92.8%. Whereas for Multiple Cameras dataset HOG model has outperformed Optical Flow model. With this dataset HOG model has shown the accuracy of 97.4% but optical flow model has shown accuracy of 96.1%. HOG model has outperformed in the cases of varying brightness across the different sections of image frame. Because of applying normalization across localized gradients during HOG feature extraction, it has neutralized the impact of varying brightness.

V. CONCLUSION AND FUTURE WORK

A new system for vision-based fall detection has been designed and implemented. The proposed system is based on Histogram of Oriented Gradients (HOG) features in a series of frames in videos. This system has been compared against the existing vision-based solution for fall detection, which is based on capturing optical flow in video frames. HOG based solution can be encouraged over optical flow based solution because HOG feature computations are relatively simple, and require less processing power while ensuring good accuracy for fall detection.

The proposed model can be improved further by including depth information, since it will help to eliminate the false cases. Also, this system of fall detection can be extended further to other applications such as injury detection after the fall by doing processing of pre and post fall scenario frames.

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